Introduce yourself in the chat!

Tell us your program and concentration

Hello there

I’m Omer, your instructor for today!

- Studied Management Science and Computer Science in my undergrad
- Graduated from the MI program at the iSchool where I studied Data Science
- Currently working for the Ontario Ministry of Children Community and Social Services as a Data Scientist
- And I love teaching!

Housekeeping

- This session is interactive! Please ask questions when prompted, I’ll be stopping regularly to check if everyone is understanding the concepts
- We will have a short break in the middle (stretch, grab some tea)
- In case, we aren’t able to cover something properly, we will skip it. The goal is to learn the content, not rush through
- We will be doing some coding together
    - Sometimes you’ll code and we then go through the solution together!
- You will have access to the slides throughout in case you need to go back to a particular slide

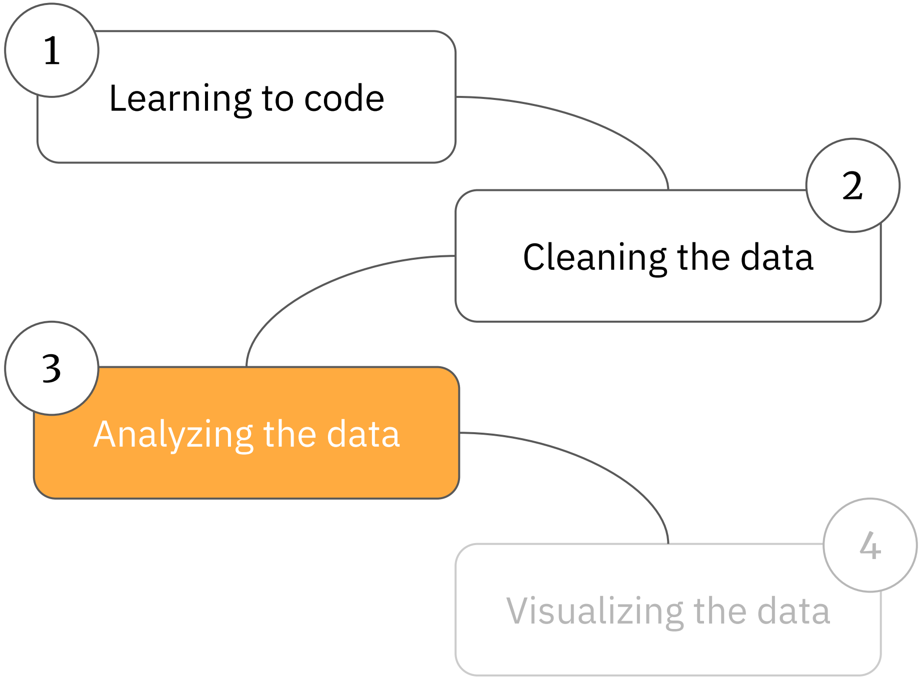
Here's What We'll Cover

1. Data storytelling
2. Exploratory data analysis
3. Recommended resources

Prologue

What we’ve studied

Our journey


1
Learning to code
2
Cleaning the data
3
Analyzing the data
4
Visualizing the data

Chapter 1

Data Storytelling

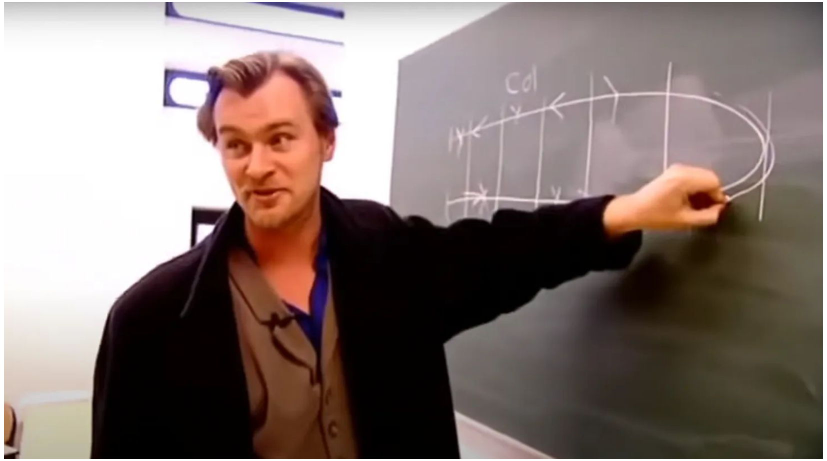
CHAPTER 1

Telling a story with data

Everyone says this but what does it really mean?

What makes a great story?

Can you think a good story? What are the key components?



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The foundation

Any good story starts with the same building blocks (Cote):

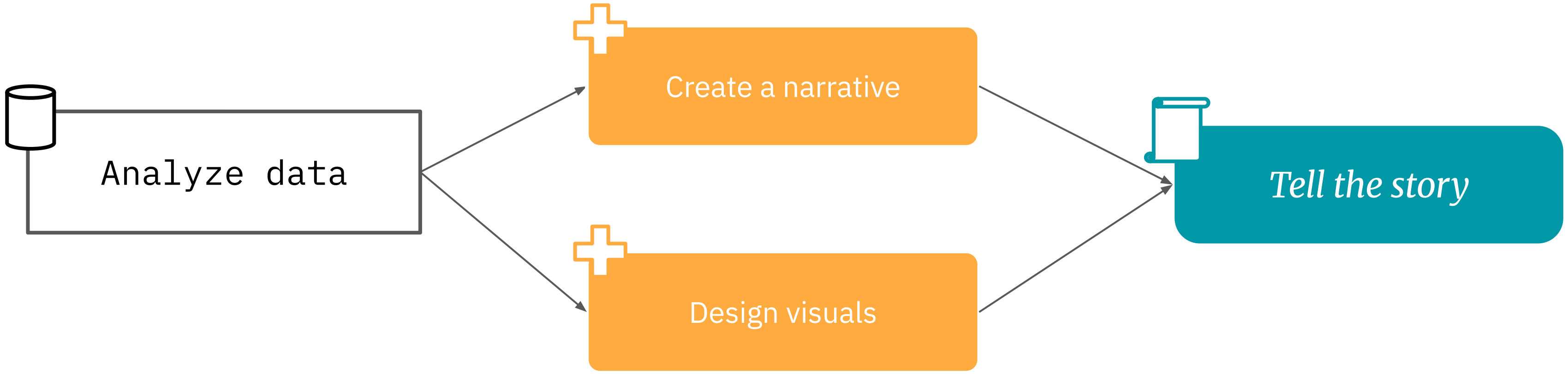
1. Characters
2. Setting
3. Conflict
4. Resolution

This applies to data storytelling as well.

Source

What is data storytelling?

It's the process of driving **insights** from **data** and communicating insights to your audience with the help of **narratives and visualization** (Cote).

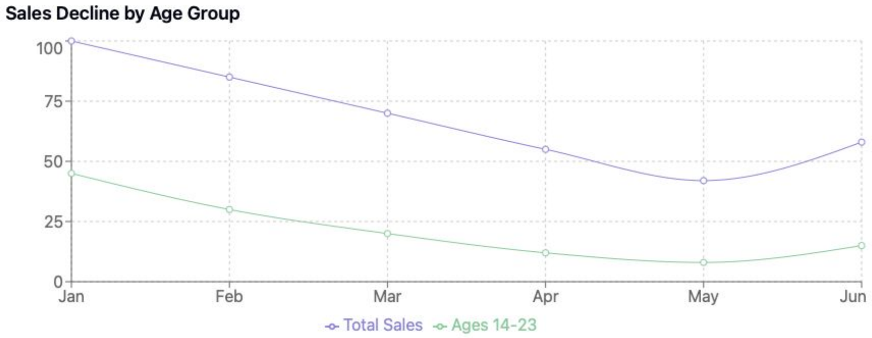


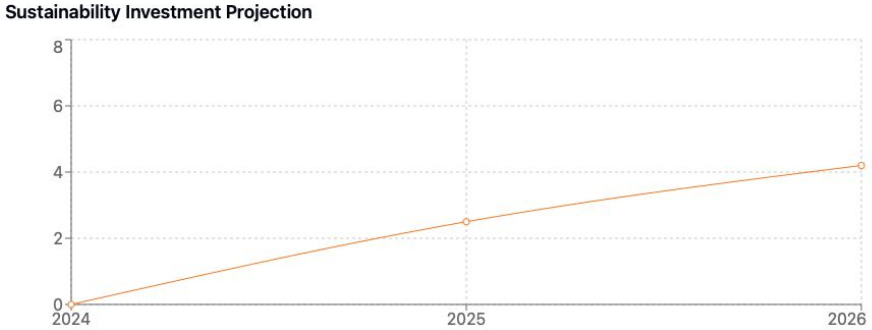
CHAPTER 1

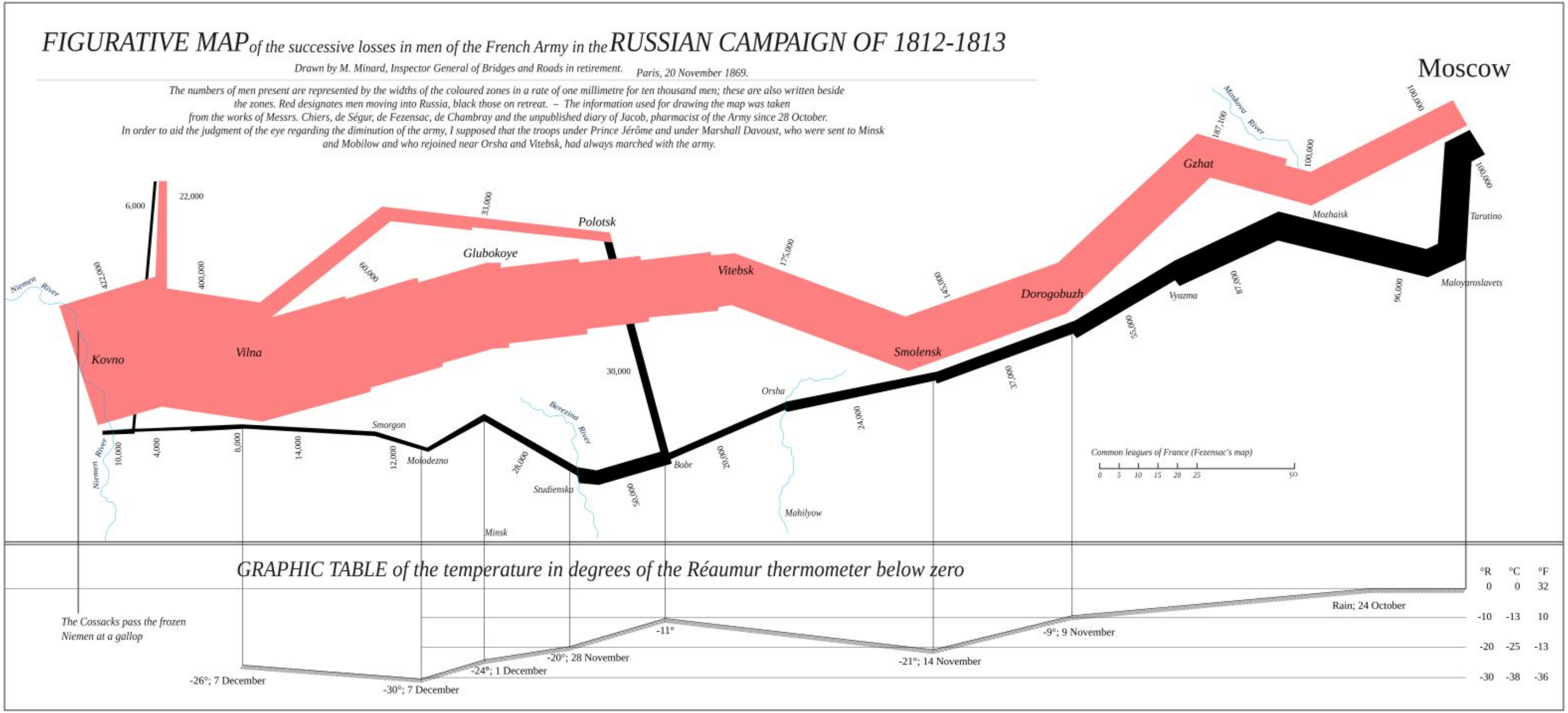
Sales are falling

We're fashion company creating trendy fast fashion clothes for Gen Z. We're receiving flak for our negative impact on the environment

- Characters
  - Customers in the younger demographic
- Setting
  - Consistent decline in sales
- Conflict
  - Social media highlighted unsustainable practices in our business
- Resolution
  - Invest in ESG projects and inform our customers







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Let’s take 5

Chapter 2

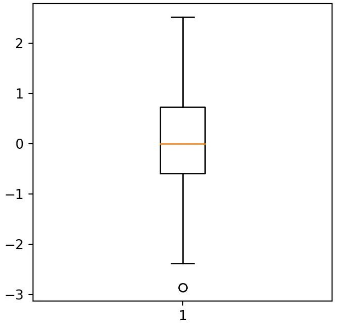
Exploratory data analysis

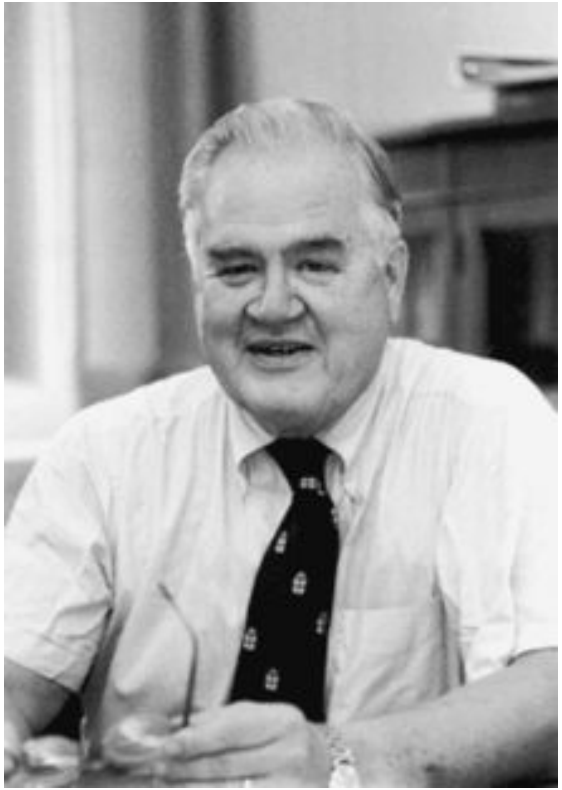
John Tukey

He is regarded as the Father of EDA.

Creator of the:

1. Boxplot
2. Tukey Test
3. and much more!





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Exploratory Data Analysis (EDA)

The goal of EDA is trying to understand your data. Our aim is understand each variable, how variables interacts with each other, and if there are any missing values (Alexander, Chapter 11)

The process can look like this:

1. Checking data types
2. Looking at parts of the dataset
3. Descriptive statistics
4. Searching for null values and duplicate values
5. Sorting, grouping, subsetting your data
6. Visualizing relationships between variables (Python 104)

df.head() | df.tail() | df.sample()

First step is to take a look at our data frame so we know what we're dealing with

- df.head(n) will give you the top n values of the data frame
- df.tail(n) will give you the bottom n values of the data frame
- df.sample(n) will give you n random rows
- head() and tail() default to 5 whereas sample defaults to 1 if n is not specified

```
df.head(3)
df.tail(10)
df.sample(2)
```

pd.unique()

This is a series method that we use to find unique values. Each column in a dataframe is a series so we can use it to find unique values for a variable

```
df["Studio"].unique()
```

df.dtypes

dtypes is an attribute of dataframes i.e. information about the dataframe. This is the ingredient label on your favourite bottle of hot sauce

- We can use df.dtypes to get a series of the data type for each column in a df
- Below is how we use it

```
df.dtypes
```

Descriptive statistics

Stat	**Definition**	**Code**
Count	Number of occurrences	value_counts()
Mean	Average (sum of all values / number of values)	mean()
Median	The middle-most value (a.k.a. 50th percentile)	median()
Min	Minimum	min()
Max	Maximum	max()
Sum	Add everything	sum()

name_of_dataframe[“Name of column”].code

For example

df[“Gross”].max()

df.describe()

You can get most of the aforementioned descriptive statistics in one method call, df.describe()

- This will skip rows with NaN values
- Can be applied to both numerical and object data types
- You can even apply it specific columns too

```
df.describe(include=”all”)
df[“Gross”].describe()
```

pd.isnull(df)

isnull() is a function in pandas that checks for null (empty) values. It returns True or False for each data point in a dataframe

- We can use it to find total number of null values in a particular variable
- We could use it find rows with null values
    - any(axis=1) below checks for null values across all variables in a row

```
pd.isnull(df)
df[df.isnull().any(axis=1)]
```

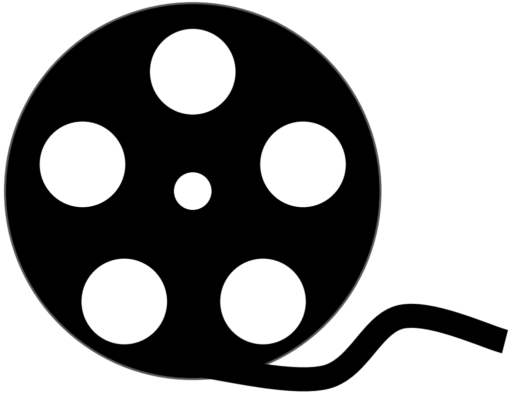
Exercise time!

We are going to be using Jupyter Notebook for our coding needs today. It’s an open-source web-application for writing computational documents. Let’s get started!

- Open jupyter.org/try
- Click on JupyterLab
- Let’s create a new notebook
    - Click on File -> New -> Notebook
    - Choose any Python Kernel
    - Click next with the default settings
    - We’re good to go!

Let's test what we've learned

We are now going to look at a dataset on movies titles by various studios. The dataset has gross revenue, gross revenue adjusted for inflation, and year too



Exercise 2.1

Time for you to practice. Using the surveys.csv file, answer the following questions

1. Do the data types of the dataframe look okay?
2. Are there any null values in the dataframe?
3. What’s the average hindfoot_length?
4. Can you list all the species_ids for me?

You have **10** minutes to do this

Let’s take 5

sort_values()

This is useful in trying to see values in ascending or descending order

name_of_dataframe.sort_values(by = “Name of column”, ascending = True or False)

For example:

df.sort_values(by= “Gross”, ascending=False)

groupby()

Grouping allows us to look at the data at an aggregate level

The value being aggregated

name_of_dataframe.groupby("Name of column")["Value"].mean()

For example:

df.groupby("Year")["Gross"].mean()

You can use other stats too!

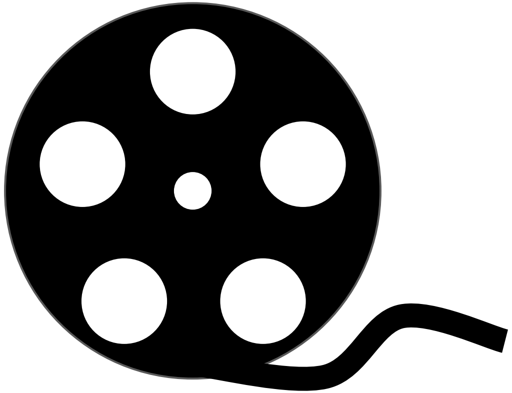
subsetting

- What if you only want to look at one decade?
- Or if you want to look at the performance of a particular studio?
- For this you'll need to filter the data with boolean indexing - filtering a dataframe with conditions

```
df[(df["Studio"] == "Sony") | (df["Studio"] == "Fox")]
df[(df["Year"] > 2000)]
```

Let’s test what we’ve learned

We shall continue fiddling with the movies dataset. This time we will play with sort_values(), groupby(), and subsetting in combination with descriptive statistics.



Exercise 2.2

Your turn again. Using the surveys.csv file, answer the following questions

1. What's the average hindfoot_length for males and females?
2. What's the median hindfoot_length for females?
3. I'm only interested in data from my birth year. What was the maximum weight in 1999?

Bonus

Recommended resources

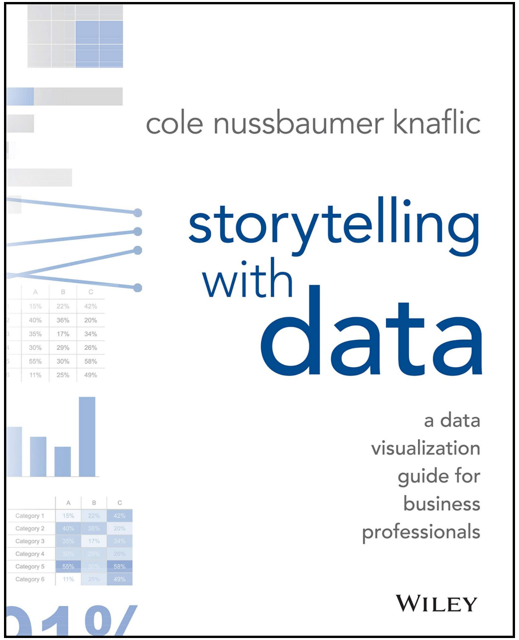
Practice!

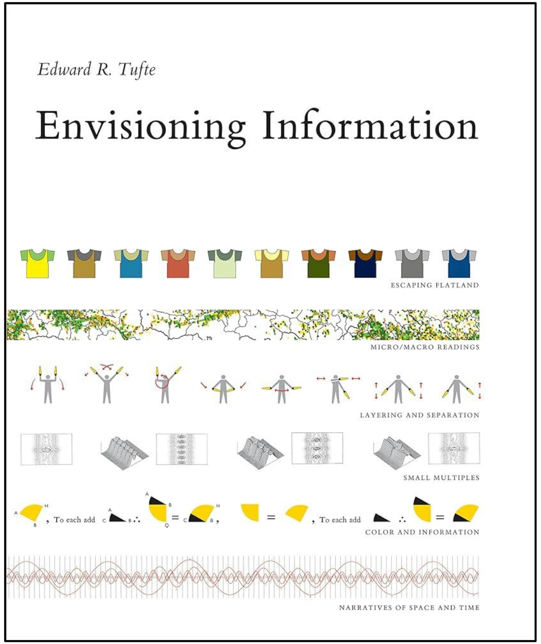
- Grab a dataset from this appendix from Rohan Alexander's book
  - Ideally on something you're passionate about
- Clean the data
- Analyze the data
- Visualize the insights
- Wrap it all together in a neat presentation or article!
- A nice article on forms of data analytics

Source

BONUS

Recommended readings





Thank you for tuning in!

I am always available for a coffee chat. Drop me a message on LinkedIn (Scan the QR code).

I would love to hear some feedback on the workshop. Please fill out this feedback survey from the iSchool.

I'll see you in Python 104!

References

Alexander, Rohan. *Telling Stories With Data*. 27 July 2023, rohanalexander.github.io/telling_stories-published

Cote, Catherine. "Data Storytelling: How to Tell a Story With Data." *Harvard Business School Online*, 23 Nov. 2021, online.hbs.edu/blog/post/data-storytelling.

pandas API Reference is frequently used to help describe how various code works

The surveys.csv file is from Portal Project Teaching Database

I try to design my content in line with my Instructor training from the Carpentries